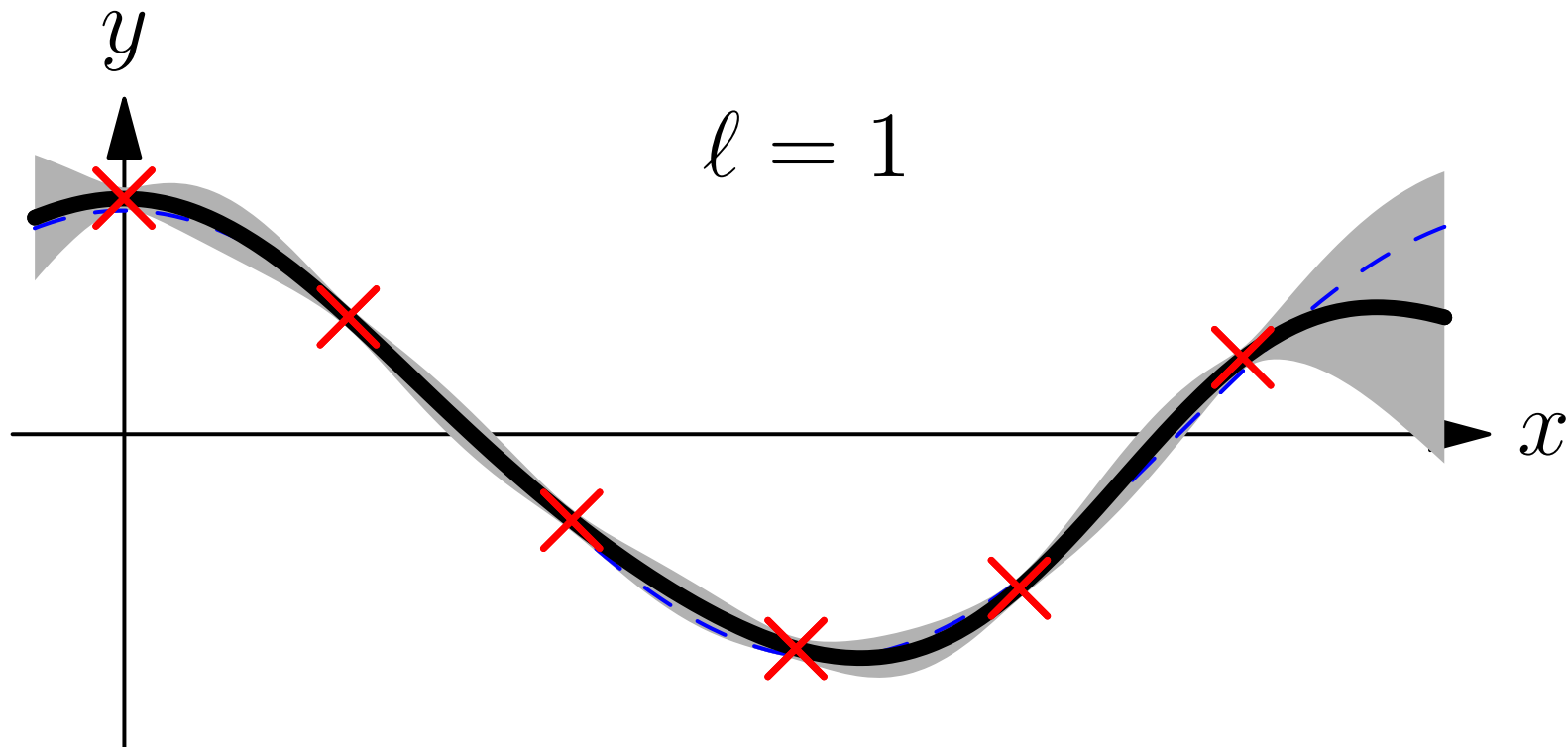


Mathematics of Machine Learning

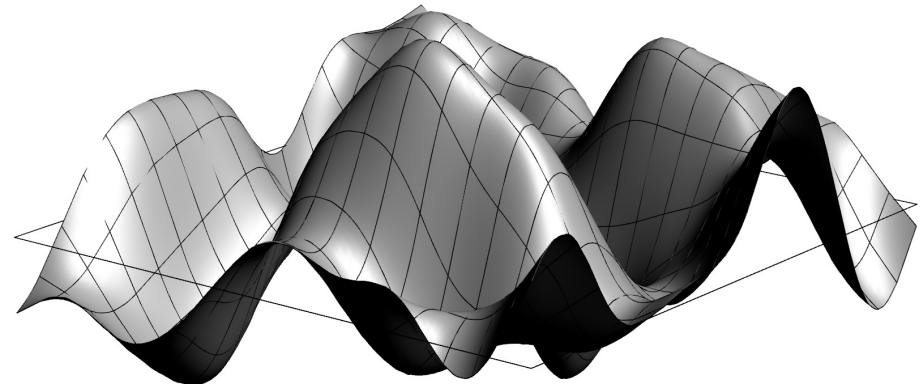
Gaussian Processes



Gaussian Processes, regression

Outline

1. **ML with Gaussians**
2. Gaussian Processes
3. Bayesian Inference
4. Hyper-parameters



Machine Learning with Gaussians

- Due to the fact that we can analytically integrate over multivariate Gaussians they are heavily used of machine learning
- Examples include multivariate modelling, Kalman filters and Gaussian Processes
- Despite being analytically tractable, they are very fiddly
- This is mainly because it is easy to specify means, μ , and covariance, $\mathbf{C}$, but Gaussians involve the inverse covariance matrix

$$\mathcal{N}(\mathbf{x} \mid \mu, \mathbf{C}) = \frac{e^{-\frac{1}{2}(\mathbf{x}-\mu)^{\top} \mathbf{C}^{-1}(\mathbf{x}-\mu)}}{\sqrt{\det(2\pi \mathbf{C})}}$$

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Completing the Square

- One of the key tools for performing Gaussian integrals is “completing the square”

$$-\frac{1}{2}\mathbf{x}^\top \mathbf{C}^{-1} \mathbf{x} + \mathbf{a}^\top \mathbf{x} = -\frac{1}{2}\mathbf{x}^\top \mathbf{C}^{-1} \mathbf{x} + \frac{1}{2}\mathbf{a}^\top \mathbf{x} + \frac{1}{2}\mathbf{x}^\top \mathbf{a}$$

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- This gives us the Hubbard–Stratonovich transformation

$$\int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}\mathbf{x}^T\mathbf{C}^{-1}\mathbf{x} + \mathbf{a}^T\mathbf{x}}}{\sqrt{|\det(2\pi\mathbf{C})|}} d\mathbf{x} = \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(\mathbf{x}^T - \mathbf{a}^T\mathbf{C})\mathbf{C}^{-1}(\mathbf{x} - \mathbf{C}\mathbf{a}) + \frac{1}{2}\mathbf{a}^T\mathbf{C}\mathbf{a}}}{\sqrt{|\det(2\pi\mathbf{C})|}} d\mathbf{x}$$

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Moment Generating Function

- Again through completing the square we can show

$$M(\mathbf{a}) = \int_{-\infty}^{\infty} e^{\mathbf{a}^\top \mathbf{x}} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{C}) d\mathbf{x} = e^{\mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{a}^\top \mathbf{C} \mathbf{a}}$$

- The first moment is given by

$$\mathbb{E}[\mathbf{x}] = \nabla_{\mathbf{a}} M(\mathbf{a}) \Big|_{\mathbf{a}=\mathbf{0}} = (\boldsymbol{\mu} + \mathbf{C} \mathbf{a}) e^{\mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{a}^\top \mathbf{C} \mathbf{a}} \Big|_{\mathbf{a}=\mathbf{0}} = \boldsymbol{\mu}$$

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$$\mathbb{E}[x_i x_j] = \frac{\partial^2 M(\mathbf{a})}{\partial x_i \partial x_j} \Big|_{\mathbf{a}=\mathbf{0}} = \mu_i \mu_j + C_{ij}$$

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$$\mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)] = C_{ij}$$

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$$C(\mathbf{a}) = \log(M(\mathbf{a})) = \mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{a}^\top \mathbf{C} \mathbf{a}$$

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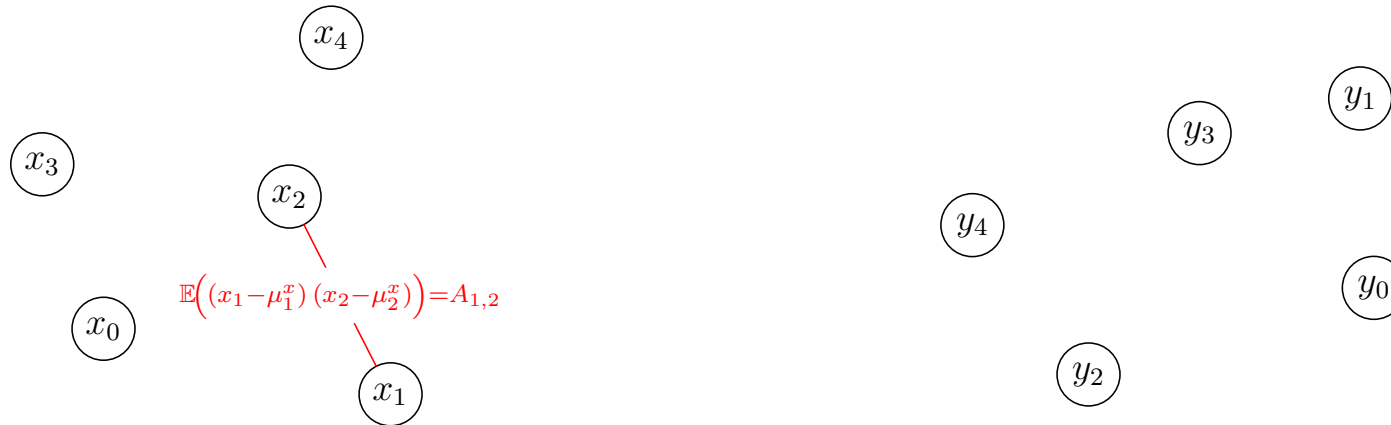
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Multivariate Normals



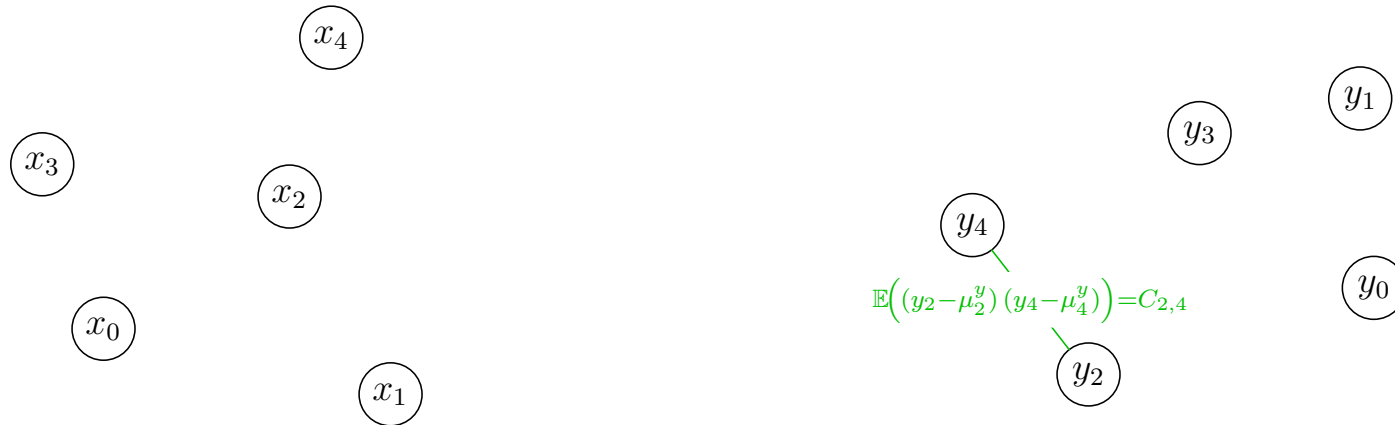
$$\mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \boldsymbol{\mu}^x \\ \boldsymbol{\mu}^y \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}\right) = \frac{e^{-\frac{1}{2}(\mathbf{x}^\top - \boldsymbol{\mu}^{x\top}, \mathbf{y}^\top - \boldsymbol{\mu}^{y\top}) \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x} - \boldsymbol{\mu}^x \\ \mathbf{y} - \boldsymbol{\mu}^y \end{pmatrix}}}{\sqrt{\det(2\pi \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix})}}$$

Multivariate Normals



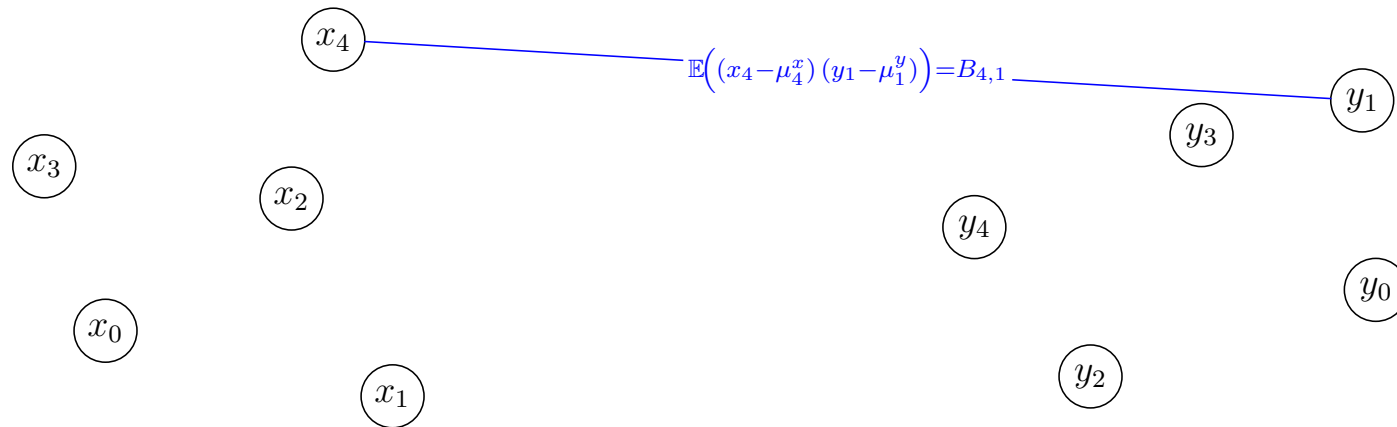
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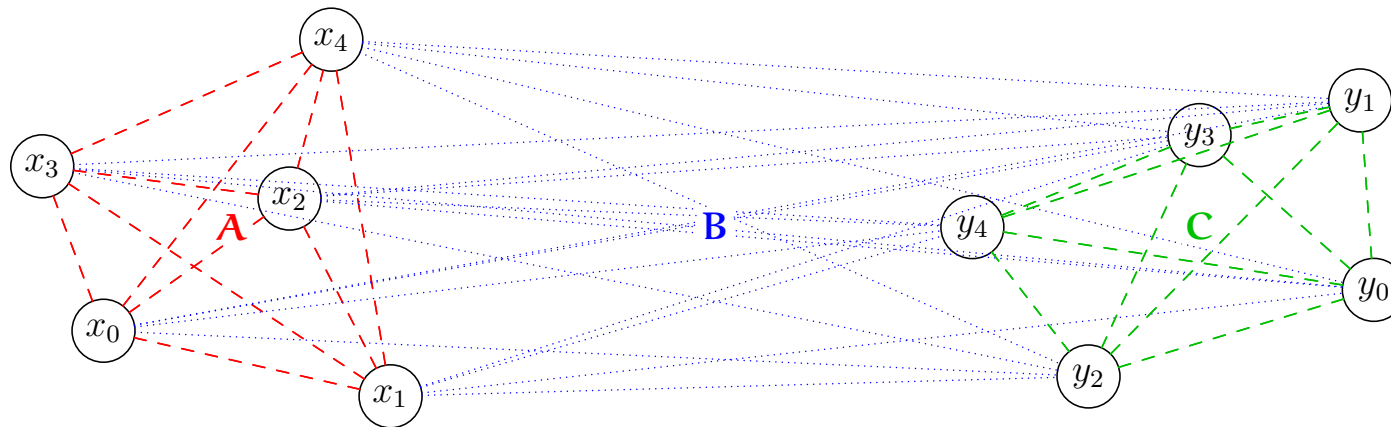
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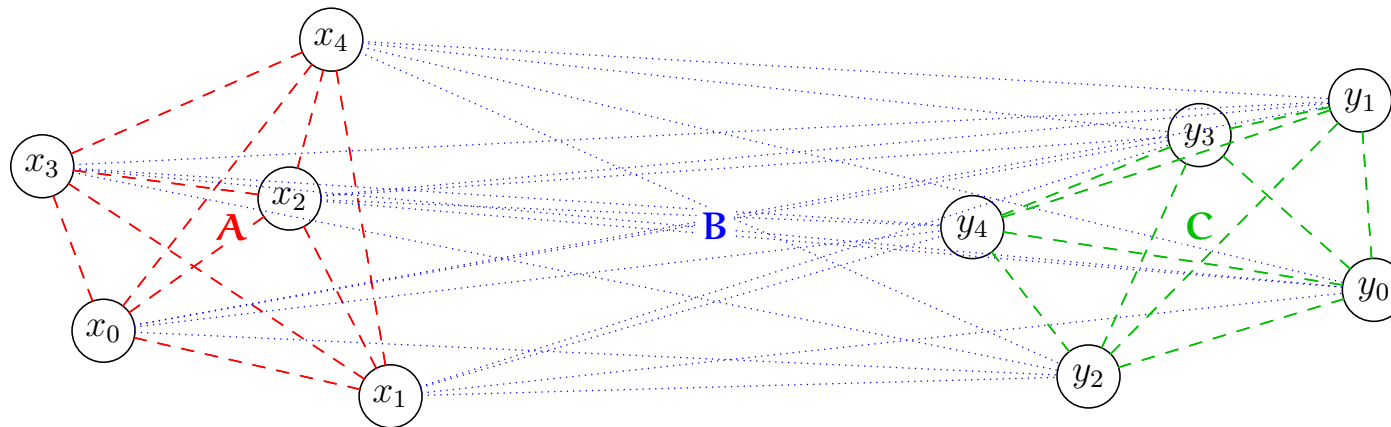
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Marginalising Multivariate Normals

- I can define a moment generating function,

$$\begin{aligned} M(\mathbf{a}, \mathbf{b}) &= \int e^{(\mathbf{a}^\top, \mathbf{b}^\top) \begin{pmatrix} \mathbf{x} - \boldsymbol{\mu}^x \\ \mathbf{y} - \boldsymbol{\mu}^y \end{pmatrix}} \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \boldsymbol{\mu}^x \\ \boldsymbol{\mu}^y \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix}\right) d\mathbf{x} d\mathbf{y} \\ &= e^{\frac{1}{2}(\mathbf{a}^\top, \mathbf{b}^\top) \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{pmatrix} \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}} = e^{\frac{1}{2}(\mathbf{a}^\top \mathbf{A} \mathbf{a} + \mathbf{b}^\top \mathbf{C} \mathbf{b} + 2\mathbf{a}^\top \mathbf{B} \mathbf{b})} \end{aligned}$$

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Marginalising Multivariate Normals Continued

- An important and non-trivial result is

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- You would think this is easy to prove by straightforward integration
- It can be done, but it is a page of work

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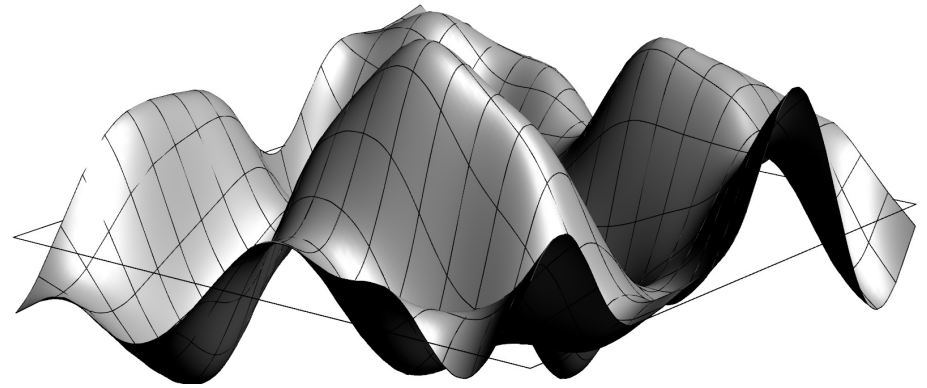
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Outline

1. ML with Gaussians
2. **Gaussian Processes**
3. Bayesian Inference
4. Hyper-parameters



Gaussian Processes

- Gaussian processes (GPs) are a mathematically defined ensemble of functions
- They can be combined with Bayesian inference to give one of the most powerful regression techniques
- Although Bayesian they can be used in a black-box fashion due to the ubiquity of the prior
- Mathematically they are a bit complicated
- In practice they aren't that difficult to use

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Regression

- In regression we try to fit a multi-dimensional function to our data
- (You can use Gaussian Processes for classification, e.g. by inferring the probabilities of being in a class, but we ignore this as regression is where GP excel)
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Priors on Functions

- We can think of a solution as a function $f(\mathbf{x})$
- We can put a prior probability distribution, $p(f)$, on a function, f , that prefers smooth functions
- We can then compute a posterior probability distribution on functions given the data, $p(f|\mathcal{D})$
- As a likelihood, $p(y_i|f(\mathbf{x}_i))$, we use the probability of observing y_i given the true function value is $f(\mathbf{x}_i)$
- In general, this would be next to impossible to compute

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Gaussian Processes

- Gaussian Processes are probability distributions over functions
- (Functions can be viewed as vectors in an infinite dimensional vector space)
- In the Gaussian Process, $\mathcal{GP}(m, k)$, the probability of a function, f , is proportional

$$p(f|m, k) \propto e^{-\frac{1}{2} \int (f(\mathbf{x}) - m(\mathbf{x})) k^{-1}(\mathbf{x}, \mathbf{y}) (f(\mathbf{y}) - m(\mathbf{y})) d\mathbf{x} d\mathbf{y}}$$

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Meaning of GP

- To understand GP's we can discretise space, $\mathbf{x}$, into a lattice of points $\{\mathbf{x}_i\}$
- Then (assuming $m(\mathbf{x}) = 0$)

$$p(f|m, k) \propto \prod_i e^{-\frac{f_i^2 k^{-1}(\mathbf{x}_i, \mathbf{x}_i)}{2} + f_i \sum_j k^{-1}(\mathbf{x}_i, \mathbf{x}_j) f_j}$$

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Covariance function

- $k(\mathbf{x}, \mathbf{y})$ is a covariance function

$$\mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{y}) - m(\mathbf{y}))] = k(\mathbf{x}, \mathbf{y})$$

- This is sometimes know as a **kernel**—it must be positive semi-definite
- It is a free “parameter” that the user gets to choose (although we can learn its parameters too)
- If $k(\mathbf{x}, \mathbf{y})$ is a function of $\mathbf{x} - \mathbf{y}$ it is “**stationary**”
- If $k(\mathbf{x}, \mathbf{y})$ is a function of $\|\mathbf{x} - \mathbf{y}\|$ it is also “**isometric**”

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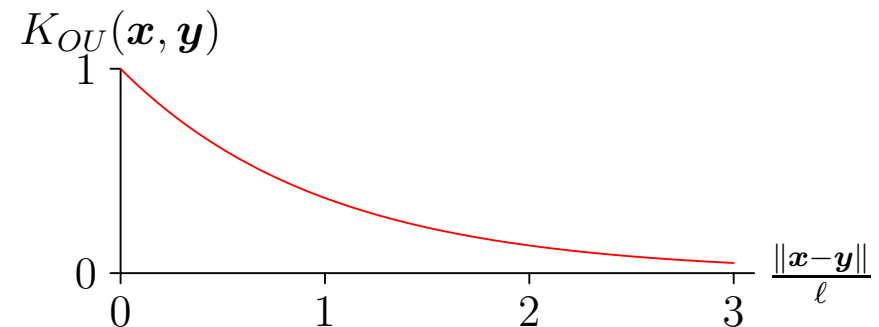
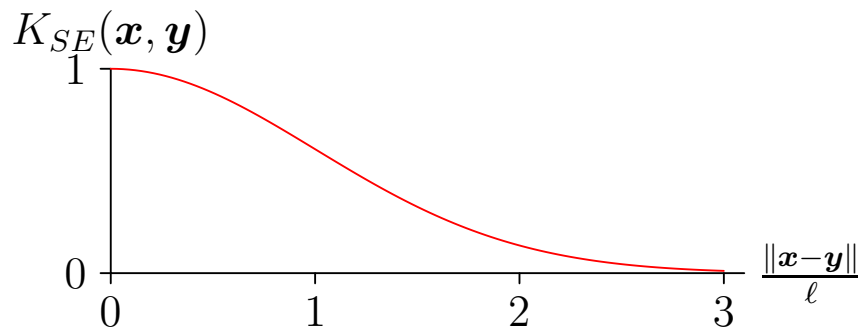
- $k(\mathbf{x}, \mathbf{y})$ is a covariance function

$$\mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{y}) - m(\mathbf{y}))] = k(\mathbf{x}, \mathbf{y})$$

- This is sometimes know as a **kernel**—it must be positive semi-definite (just like in SVMs)
- It is a free “parameter” that the user gets to choose (although we can learn its parameters too)
- If $k(\mathbf{x}, \mathbf{y})$ is a function of $\mathbf{x} - \mathbf{y}$ it is “**stationary**”
- If $k(\mathbf{x}, \mathbf{y})$ is a function of $\|\mathbf{x} - \mathbf{y}\|$ it is also “**isometric**”

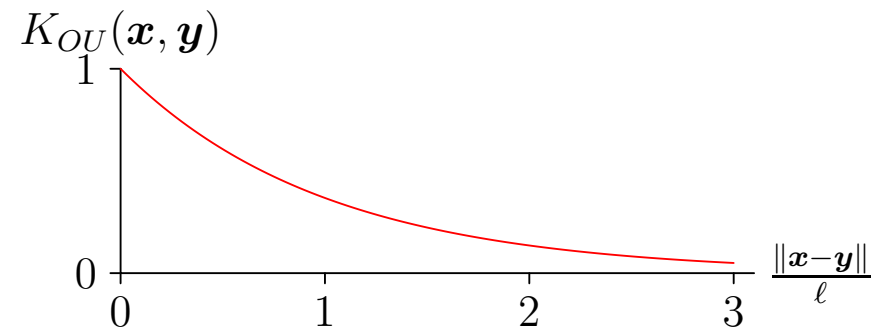
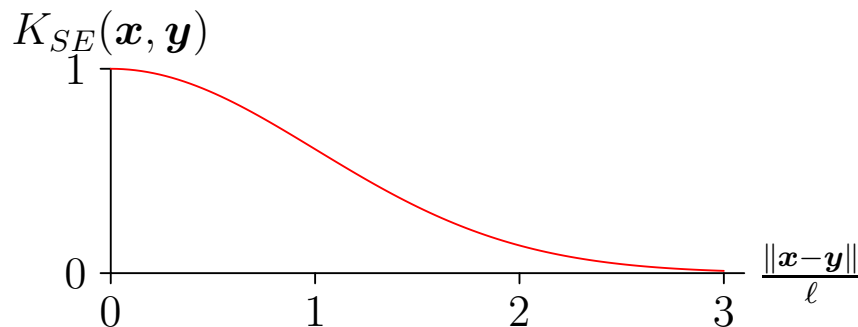
Popular Choices of GP Kernel Function

- Constant: $k_C(\mathbf{x}, \mathbf{y}) = C$
- Gaussian noise: $k_{\text{GN}}(\mathbf{x}, \mathbf{y}) = \sigma^2 \delta_{\mathbf{x}, \mathbf{y}}$
- Squared exponential: $k_{\text{SE}}(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{2\ell^2}\right)$
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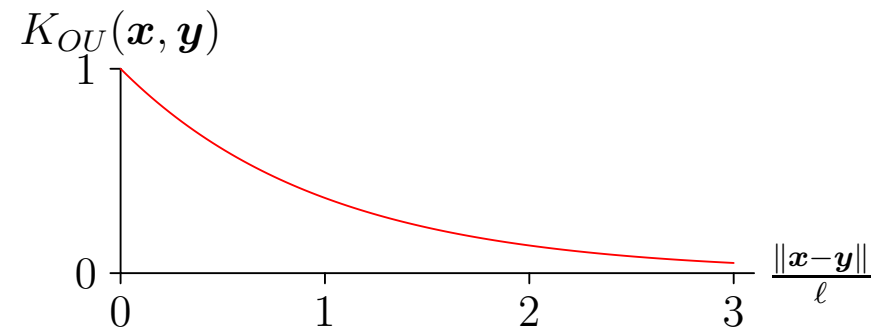
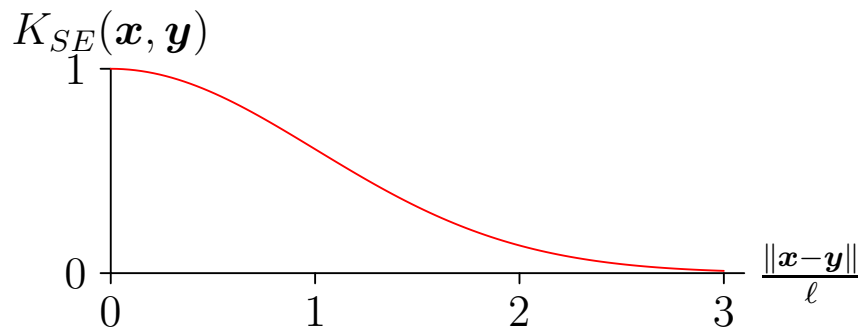
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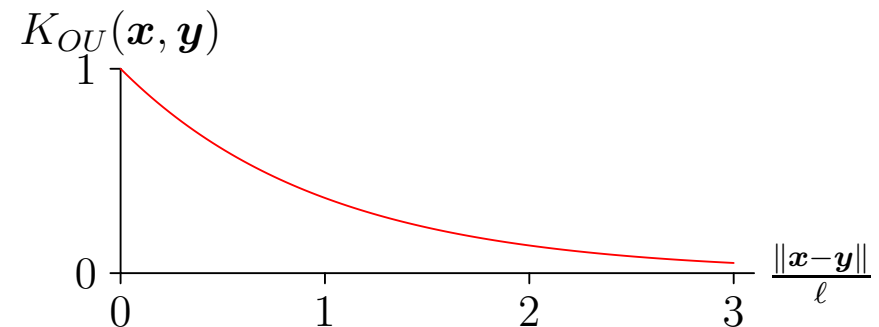
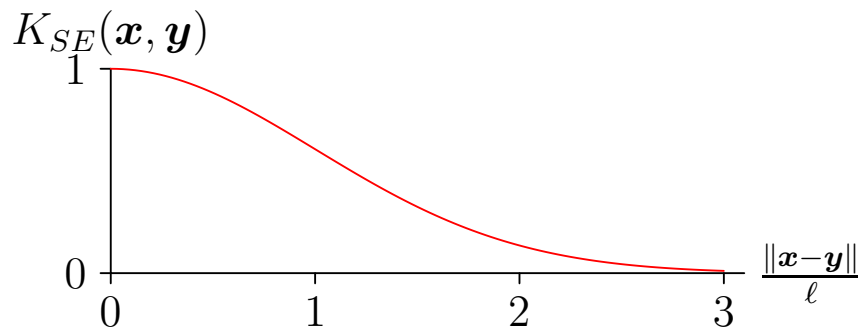
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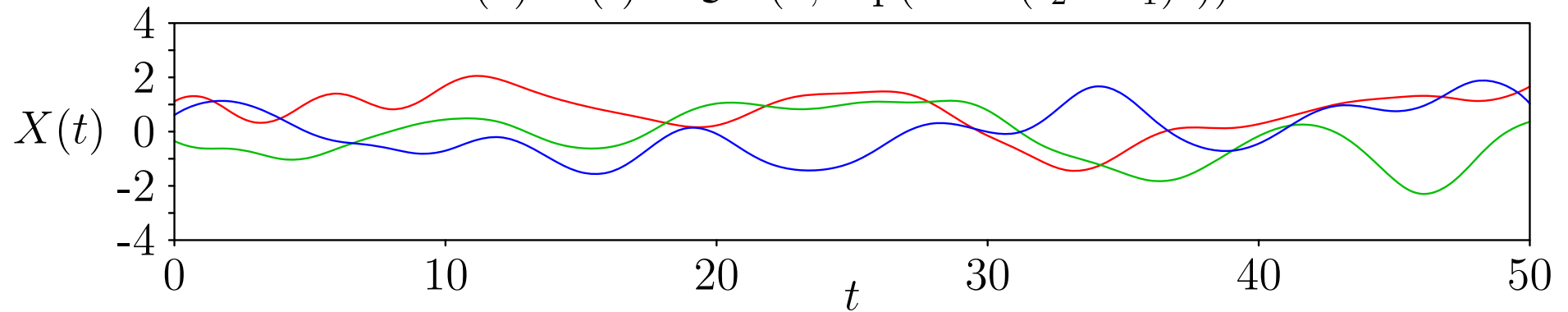
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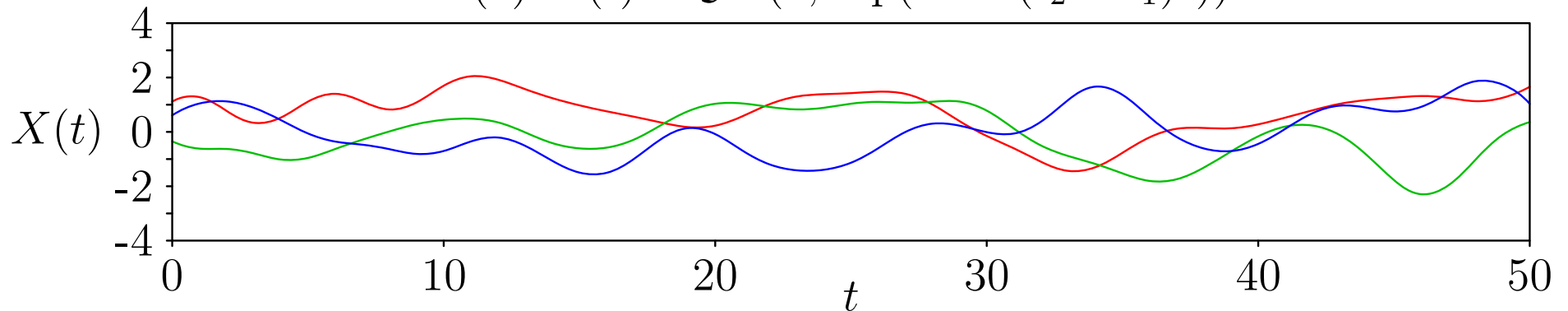
Gaussian Process Worlds

(a) $X(t) \sim \mathcal{GP}(0, \exp(-0.1(t_2 - t_1)^2))$

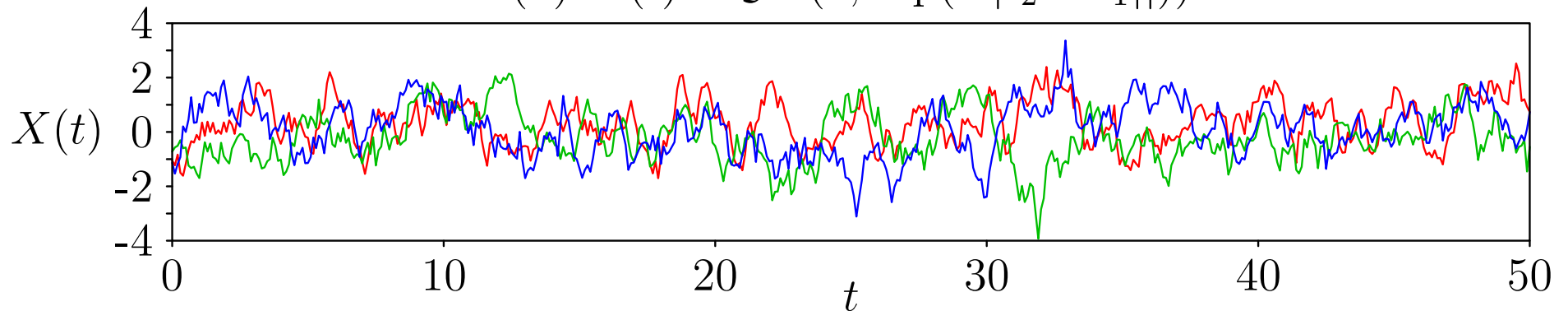


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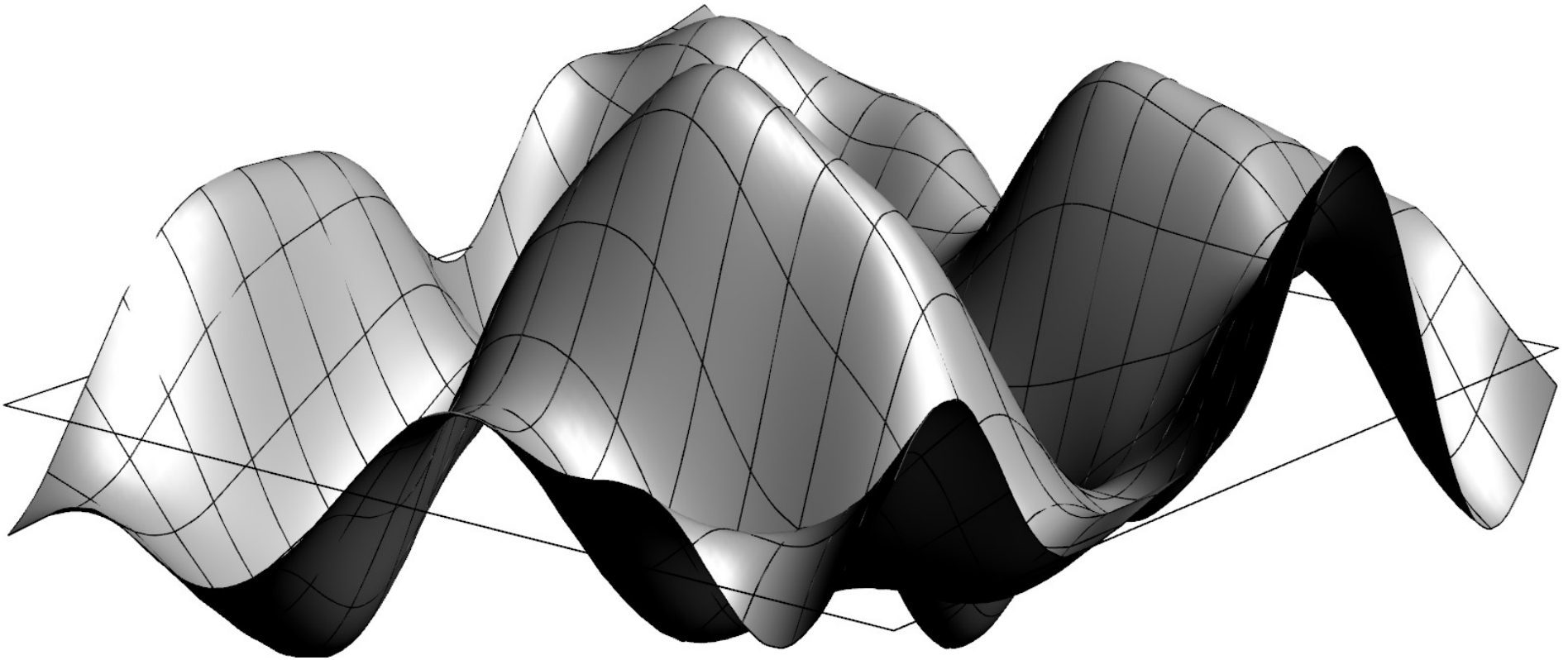
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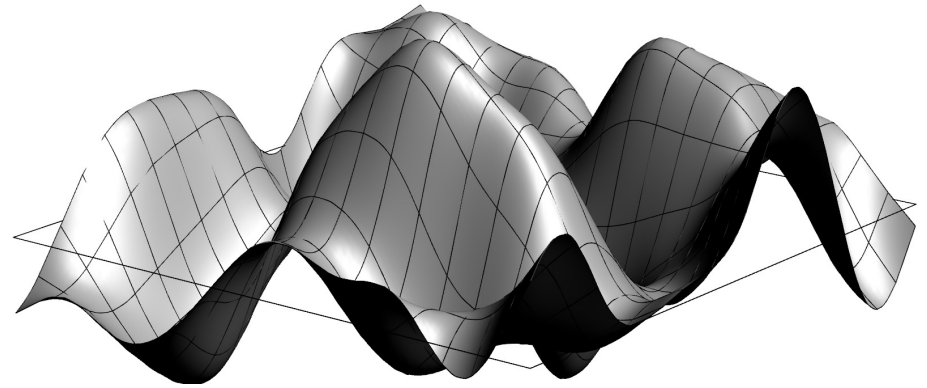


2-D Gaussian Processes



Outline

1. ML with Gaussians
2. Gaussian Processes
3. **Bayesian Inference**
4. Hyper-parameters



Observed Gaussian Processes

- Given some data points $\mathcal{D} = ((\mathbf{x}_i, y_i) | i = 1, \dots, m)$ the likelihood (assuming Gaussian error are independence of the data point) is given by

$$p(\mathcal{D}|f) = \prod_{i=1}^m \mathcal{N}(y_i | f(\mathbf{x}_i), \sigma^2)$$

- Using a Gaussian Process prior we can compute a posterior using Bayes's rule
- The posterior is a Gaussian Process with a shifted mean and variance depending on the data-points
- This direct Bayesian derivation gives the answer involving the inverse matrix of the correlation function, $k^{-1}(\mathbf{x}, \mathbf{y})$

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Alternative Derivation

- Denoting the target values as a vector $\mathbf{y}$ with elements y_i
- Denoting the matrices of covariances between data points as $\mathbf{K}$ with elements $k(\mathbf{x}_i, \mathbf{x}_j)$
- Denoting the covariance between the data points and a particular position, $\mathbf{x}_*$ as $\mathbf{k}_*$ with elements $k(\mathbf{x}_i, \mathbf{x}_*)$
- Denoting the variance a point $\mathbf{x}_*$ as $k_* = k(\mathbf{x}_*, \mathbf{x}_*)$
- Then the distribution of function values at points at $\mathbf{x}_i$ and $\mathbf{x}_*$ is

$$p(\mathbf{y}, f_*) = \mathcal{N}\left(\begin{pmatrix} \mathbf{y} \\ f_* \end{pmatrix} \middle| \mathbf{0}, \begin{pmatrix} \mathbf{K} + \sigma^2 \mathbf{I} & \mathbf{k}_* \\ \mathbf{k}_*^\top & k_* \end{pmatrix}\right)$$

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Conditional Probability

- To compute the posterior $p(f_*|\mathbf{y})$ we use

$$p(f_*|\mathbf{y}) = \frac{p(f_*, \mathbf{y})}{p(\mathbf{y})}$$

- where $p(\mathbf{y}) = \int p(f_*, \mathbf{y}) \mathrm{d}f_*$
- Because all integrals are Gaussian we can compute the integral to obtain

$$p(f_*|\mathbf{y}) = \mathcal{N}\left(f_* \left| \mathbf{k}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}, k - \mathbf{k}_*^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_* \right.\right)$$

- Looks complicated, but numerically easy to evaluate

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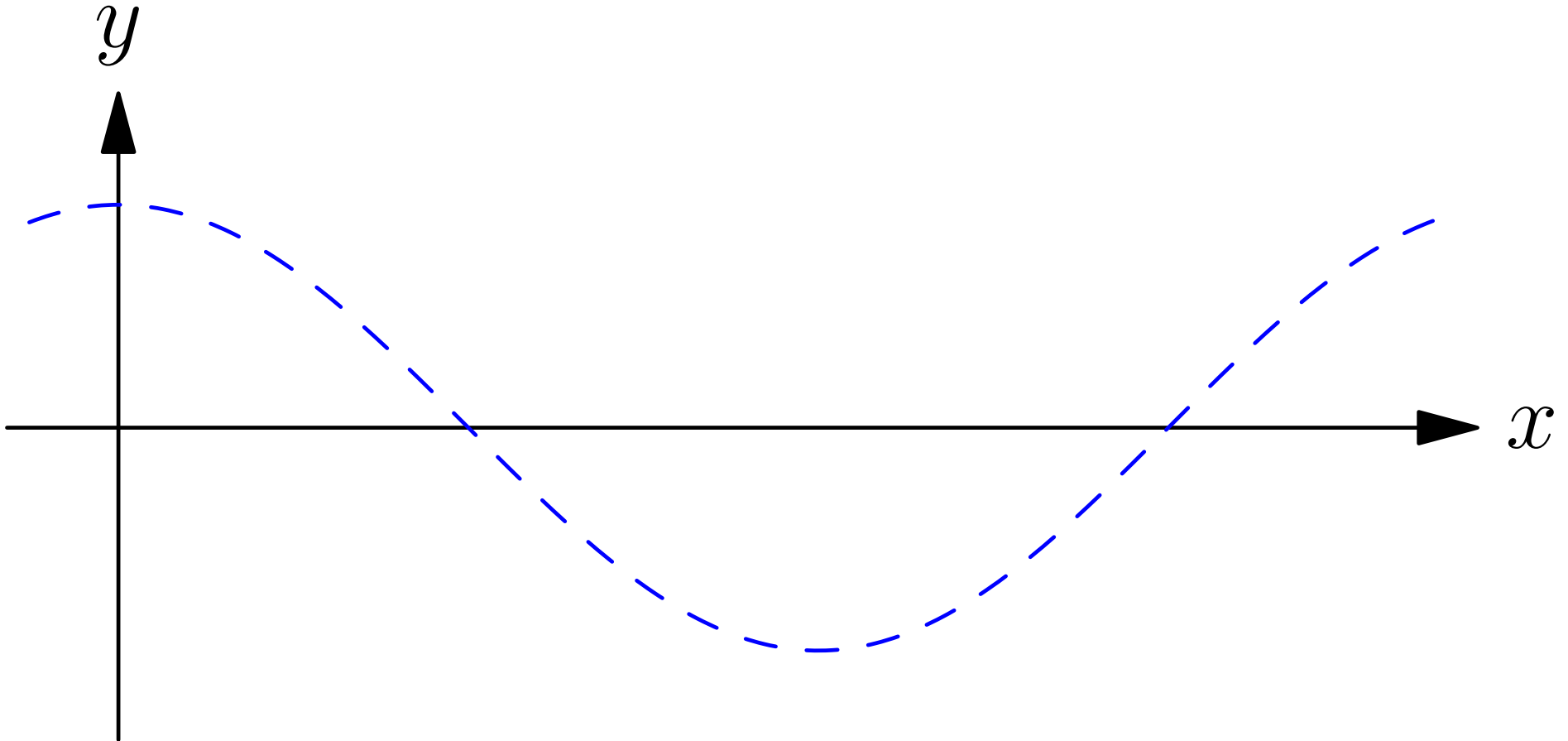
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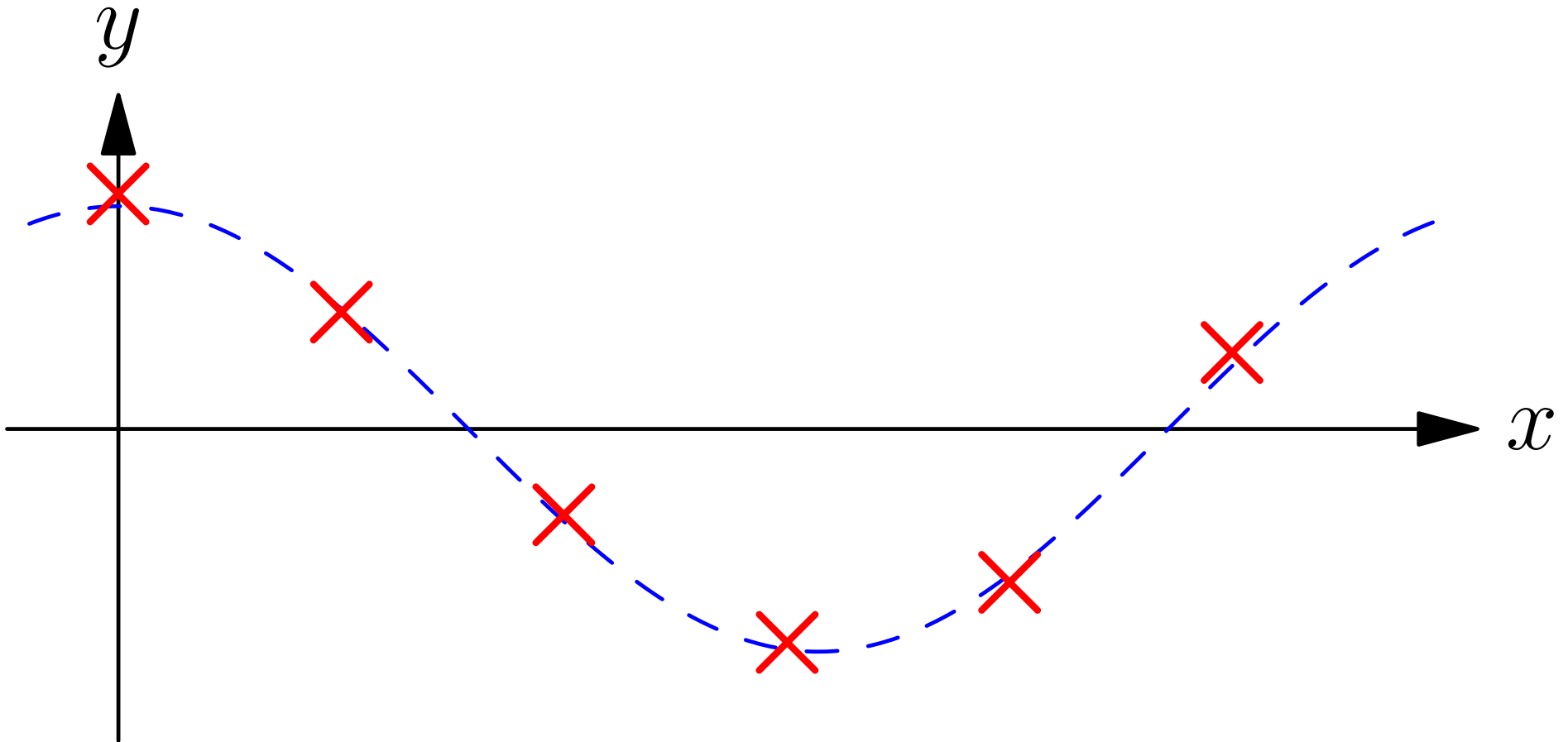
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$$K(x, x') = \exp(-(x - x')^2 / (2\ell^2))$$

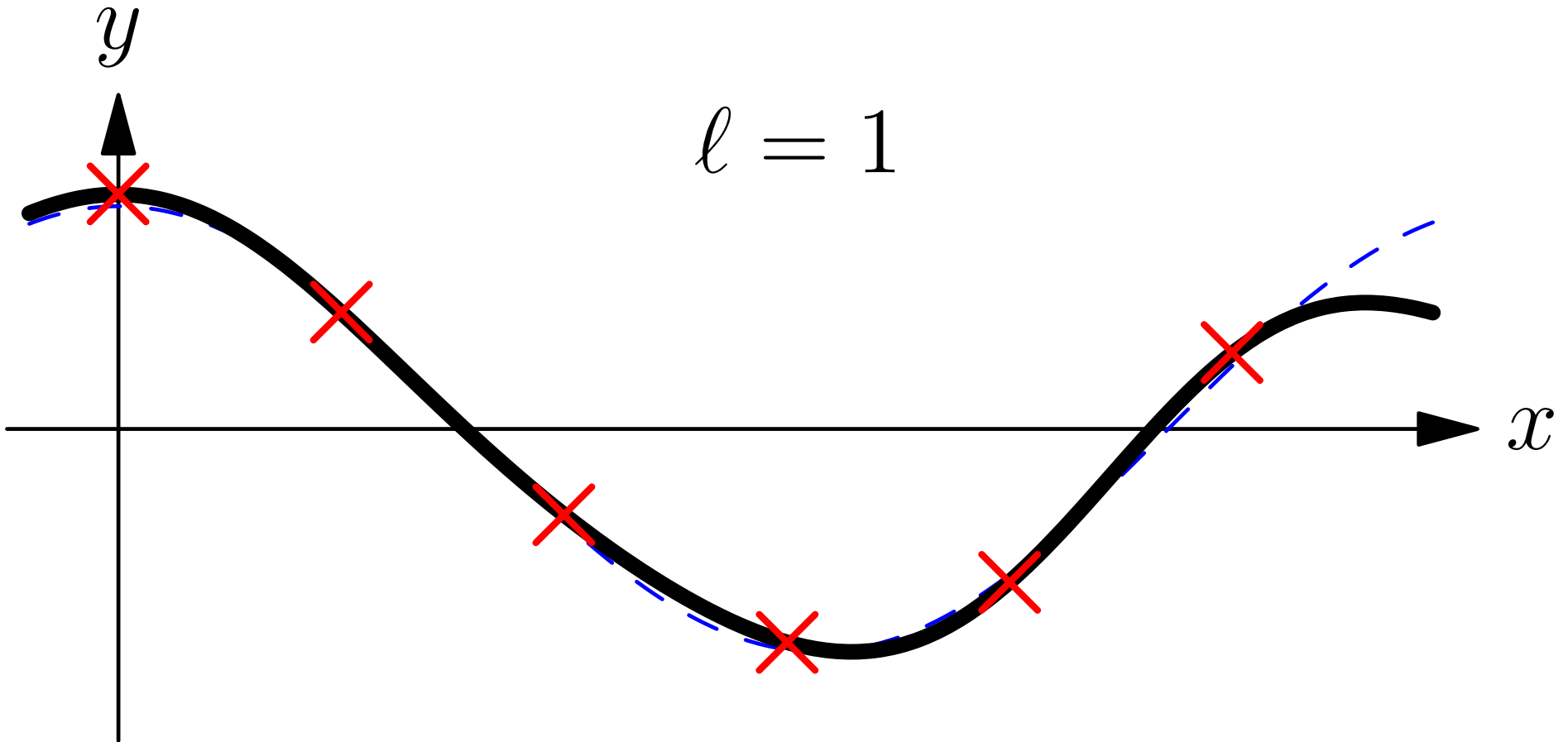


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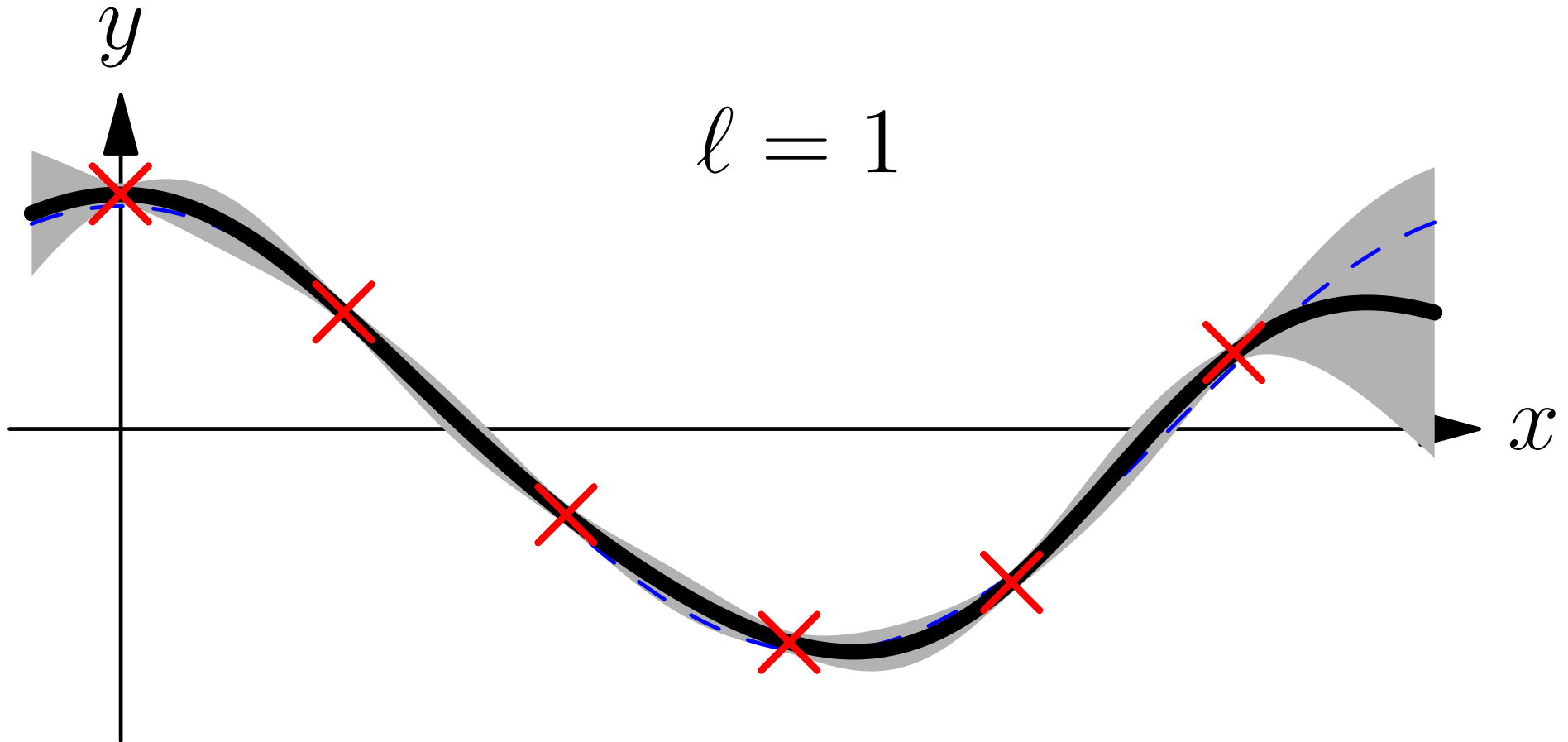
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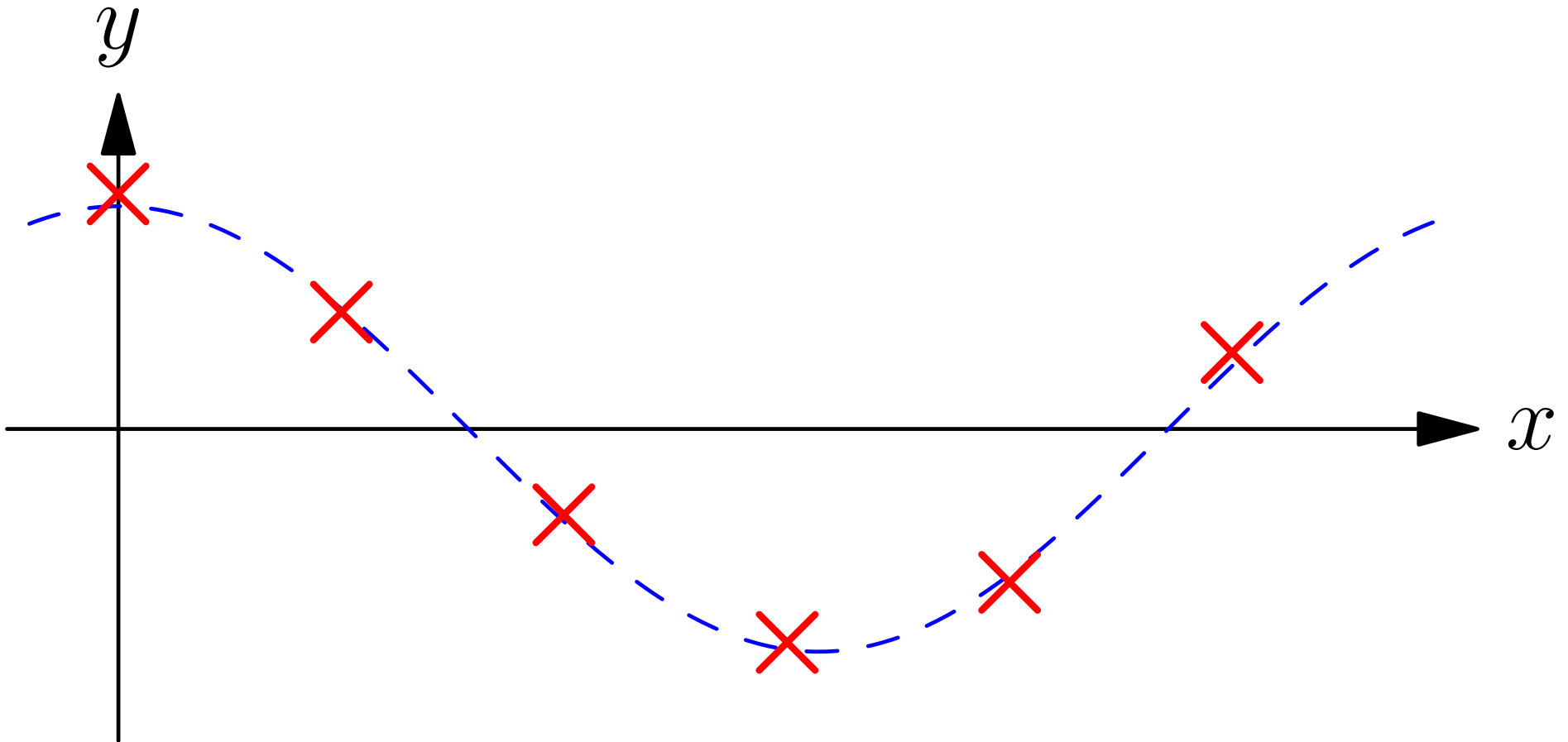


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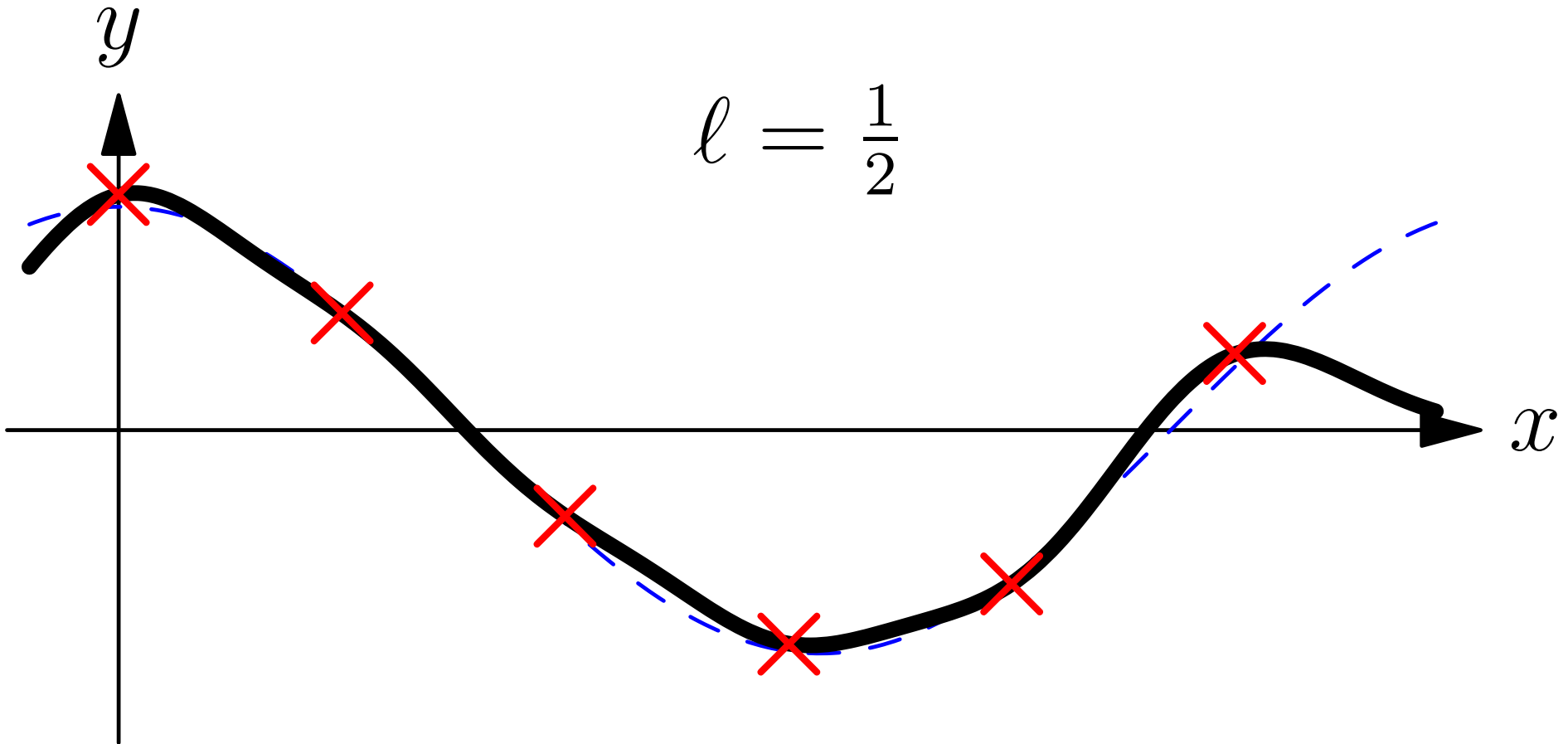


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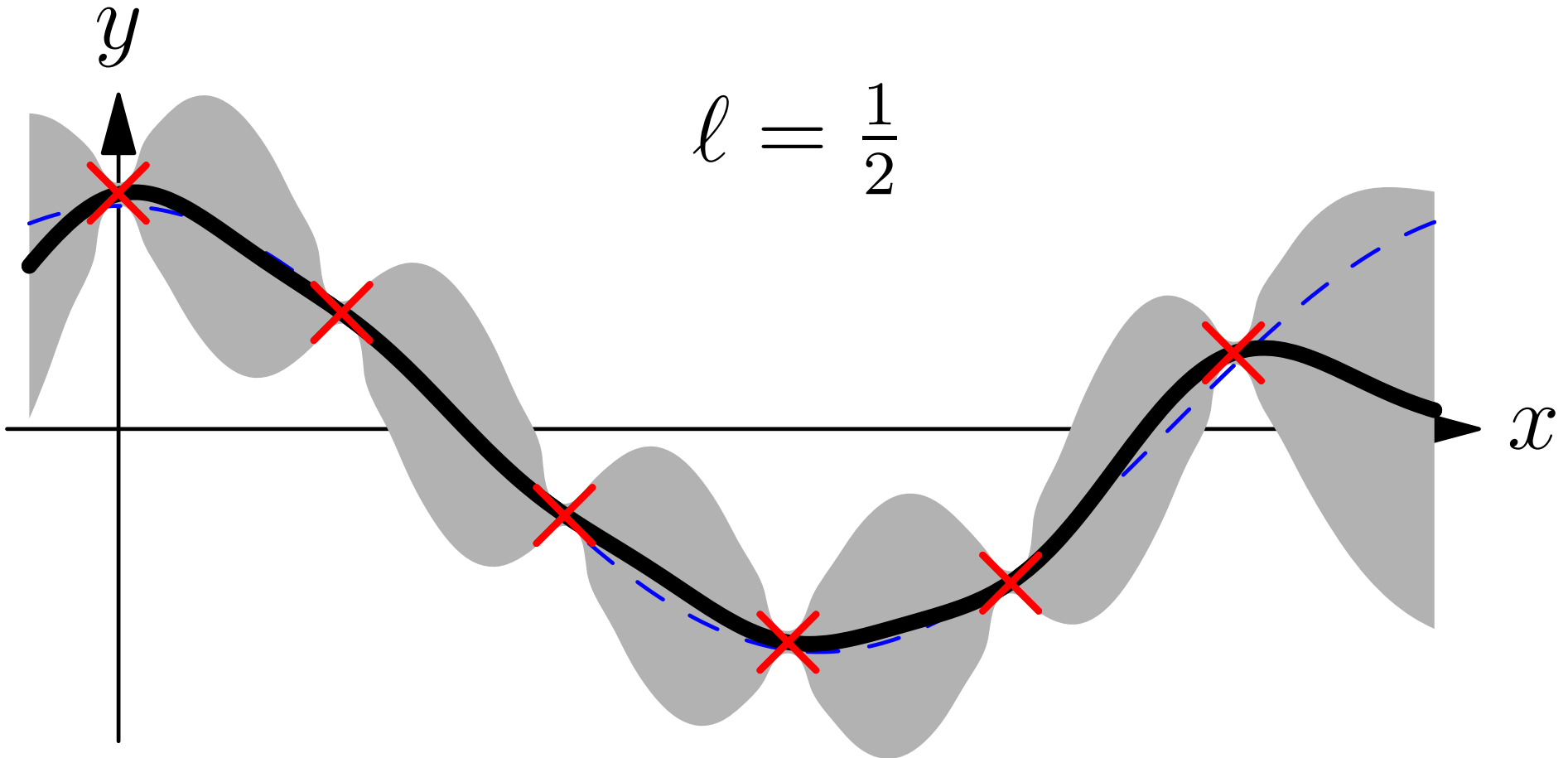
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Multi-dimensional Regression

- I've shown a 1-D regression example because it is easy to visualise
- This might be used with a time series
- The much more typical situation in machine learning is for x to have many features so we are doing multi-dimensional regression
- Gaussian process inference were first used in spatial problems where it was known as **kriging**
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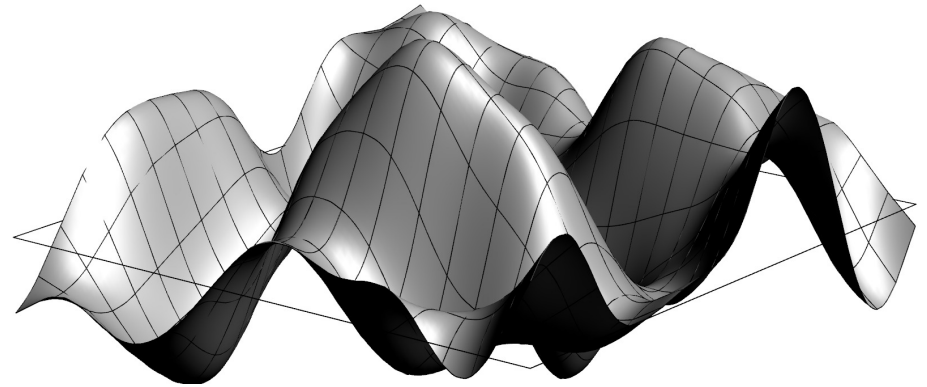
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Outline

1. ML with Gaussians
2. Gaussian Processes
3. Bayesian Inference
4. **Hyper-parameters**



Choosing the Correct Covariance Function

- Choosing the correct covariance function is critical
- Most covariance functions include a continuous **hyper-parameter** (e.g. the correlation length ℓ) that we have to choose correctly
- This is typical of many Bayesian problems where we have some set of hyper-parameters, ϕ , describing the model
- These are different to the normal parameters we learn (e.g. weights w or in GP the functions $f(x)$)
- In Bayesian inference we learn the posterior for these normal parameters

$$p(f|\mathcal{D}, \phi) = \frac{p(\mathcal{D}|f, \phi)p(f|\phi)}{p(\mathcal{D}|\phi)}$$

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Evidence Framework

- The normalisation factor, $p(\mathcal{D}|\phi)$ is known as the **marginal likelihood** or **evidence**

$$p(\mathcal{D}|\phi) = \int p(\mathcal{D}|f, \phi) p(f|\phi) df$$

- We can perform a Bayesian calculation at a second level by putting a prior on ϕ

$$p(\phi|\mathcal{D}) = \frac{p(\mathcal{D}|\phi)p(\phi)}{p(\mathcal{D})}$$

- From this we can now marginalise out the hyper-parameters

$$p(f|\mathcal{D}) = \int p(f|\mathcal{D}, \phi) p(\phi|\mathcal{D}) d\phi$$

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Maximum-Likelihood-II

- The integral

$$p(f|\mathcal{D}) = \int p(f|\mathcal{D}, \phi) p(\phi|\mathcal{D}) d\phi$$

usually can't be computed analytically and we have to use Monte Carlo methods (see later lecture)

- An alternative is to use the most likely hyper-parameter
- We can find this by using gradient search of $p(\mathcal{D}|\phi)$
- This is sometimes referred to as ML-II
- Normally even this can be difficult, but for GP its not too difficult

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usually can't be computed analytically and we have to use Monte Carlo methods (see later lecture)

- An alternative is to use the most likely hyper-parameter
- We can find this by using gradient search of $p(\mathcal{D}|\phi)$
- This is sometimes referred to as ML-II
- Normally even this can be difficult, but for GP its not too difficult

Maximum-Likelihood-II

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Evidence for GP

- For GP the (log)-evidence can be computed in closed form

$$\log(p(\mathcal{D}|\phi)) = -\frac{1}{2}\mathbf{y}^\top(\mathbf{K} + \sigma^2\mathbf{I})\mathbf{y} - \frac{1}{2}\log(|\mathbf{K} + \sigma^2\mathbf{I}|) - \frac{m}{2}\log(2\pi)$$

- ★ First term measures goodness of fit
 - ★ Second term measure complexity of model
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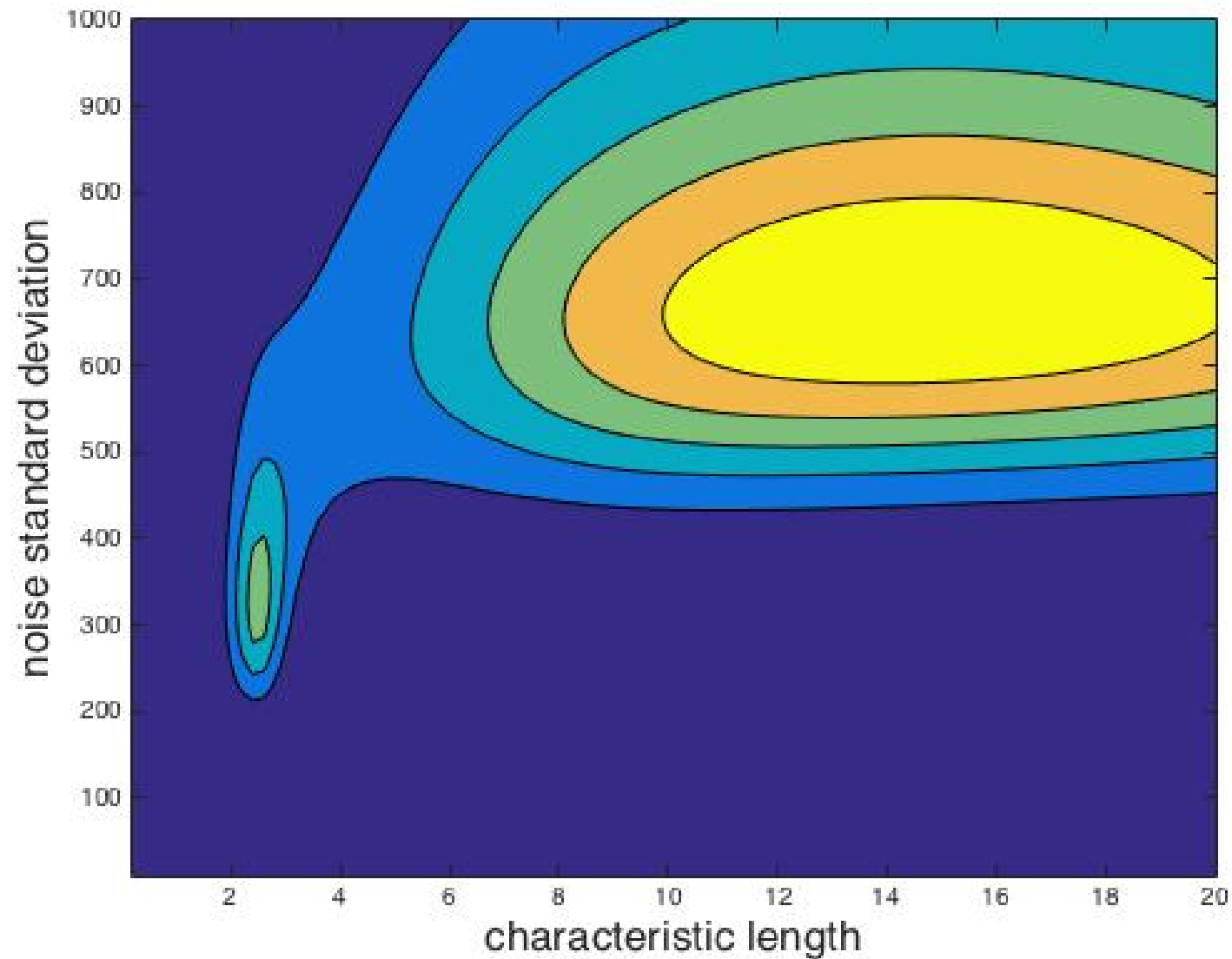
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 - Could overfit!

Example (slightly pathological)



Conclusions

- Gaussian processes are very powerful for regression (and classification?)
- Because all calculations involve Gaussian integrals we can compute everything in closed form
- (Actually its a pain to do the mathematics because you end up working with inverse of matrices)
- Fairly generic (black-box) technique because the prior captures many continuity constraints
- We can use the evidence framework (probability of data) to do model selection and hyper-parameter optimisations

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